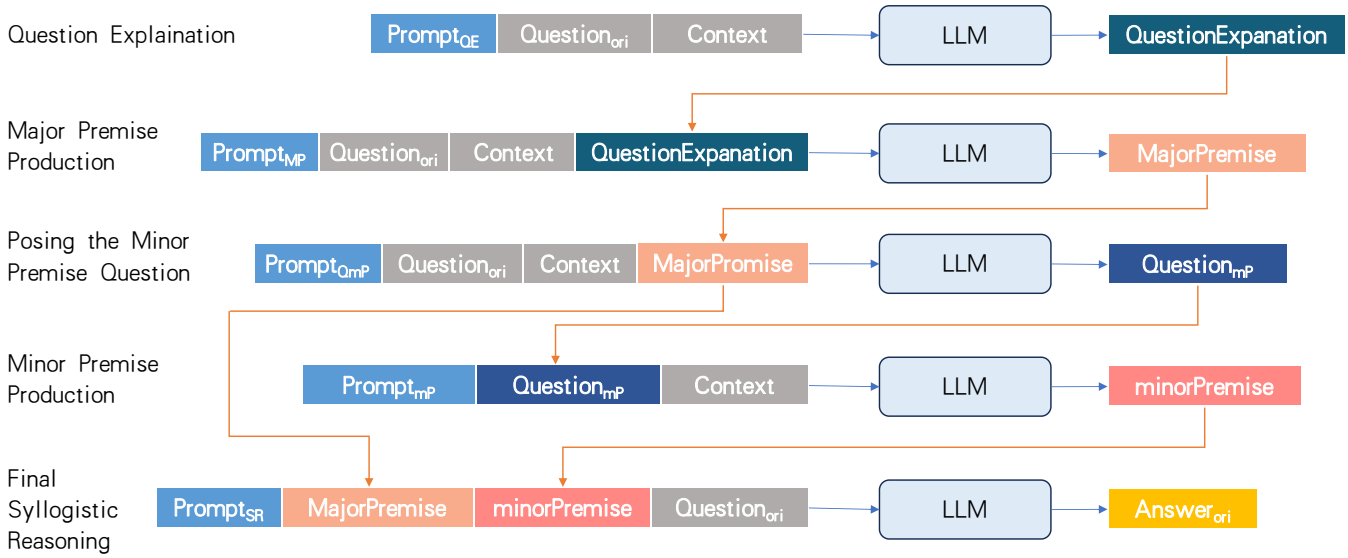

Figure 2: Procedure of our SR-FoT. Question$_{ori}$: Original Question, Context: Context provided for Original Question, Answer$_{ori}$: Answer for Original Question, Question$_{mP}$: Question for Minor Premise, Propmt$_{CoT}$: Guide Prompt for CoT, Propmt$_{QE}$: Guide Prompt for Question Explanation, Propmt$_{MP}$: Guide Prompt for Major Premise Production, Propmt$_{QmP}$: Guide Prompt for Posing the Minor Premise Question, Prompt$_{mP}$: Guide Prompt for Minor Premise Production, Prompt$_{SR}$: Guide Prompt for Final Syllogistic Reasoning and so on.

Method	GPT-3.5-turbo			DeepSeek-V2			Qwen1.5-32B-Chat		
	ScienceQA	**StrategyQA**	**BoolQ**	**ScienceQA**	**StrategyQA**	**BoolQ**	**ScienceQA**	**StrategyQA**	**BoolQ**
Single-Round									
Base	85.9	84.8	79.4	87.0	92.3	86.3	87.4	90.8	83.4
CoT (Wei et al. 2022)	86.9	91.3	79.4	88.6	91.8	86.0	87.1	91.0	83.0
SR-FoT (Ours)	**87.4**	**92.1**	**81.7**	**91.8**	**93.2**	**87.3**	**88.7**	**91.9**	**84.6**
Multi-Round									
SC-CoT (Wang et al. 2022)	87.4	91.5	82.8	88.6	92.4	87.4	87.6	92.3	84.3
C-CoT (Fu et al. 2022)	87.4	90.2	81.0	88.5	92.3	87.4	87.2	92.0	83.9
SC-SR-FoT (Ours)	**88.9**	**93.4**	**85.0**	**93.0**	**95.0**	**89.4**	**90.4**	**94.4**	**88.2**

Table 1: Comparison with the state-of-the-art methods on the ScienceQA, StategyQA, and BoolQ datasets.

choice science questions. It can be divided into three subjects: natural science, language science, and social science. It requires the language model to select one answer from multiple options, usually requiring multi-step reasoning. In our experiment, we employ the test set samples which only have a text context, with a total of 2224. We report the accuracy of our SR-FoT and comparison methods on this set.

StrategyQA (Geva et al. 2021) is a question-answering dataset focusing on open-domain questions. Its questions contain multiple reasoning steps, and a strategy should be used to obtain the answers. In our experiment, we evaluate the methods with accuracy on the train set, which includes 2290 samples.

BoolQ (Clark et al. 2019) is a reading comprehension dataset consisting of 16k samples. They often query for complex, non-factoid information, and require difficult entailment-like inference to solve. In our experiment, we compare the accuracy of our SR-FoT with other methods on the dev set, with a total of 3270.

Experimental Setting. Our experiments are performed using API calls on the proprietary model GPT-3.5-turbo (Ouyang et al. 2022), the open-source model DeepSeek-v2 (DeepSeek-AI 2024) with 236B parameters, and Qwen1.5-32B-Chat (Bai et al. 2023) version with 32B parameters. The control group methods include the Base method, Chain of Thought (CoT) (Wei et al. 2022), Self-consistency CoT (SC-CoT) (Wang et al. 2022), and Complexity-based CoT (C-CoT) (Fu et al. 2022) methods. Our own approaches included SR-FoT and Self-consistency SR-FoT (SC-SR-FoT), which represents our SR-FoT following the self-consistency sampling and aggregation settings of SC-CoT. In the single-round sampling methods which include Base, CoT and SR-FoT, the hyperparameters on GPT-3.5-turbo and Qwen1.5-32B-Chat are set to top_p=0.3 and temperature=0.2, while the temperature on DeepSeek-v2 was set to the default recommended value of 1 (DeepSeek only allows the temperature hyperparameter to be adjusted). In the multi-round sampling methods which in-

explain_prompt = """You are a knowledgeable scholar. Below is a question with corresponding context. Please read and understand the question carefully, explain the meaning of the question, and find what information in the context is needed to answer this question. Please note that the information you find must be from the context, and you cannot directly answer this question and assume some information does not appear in the context.

Question: {question}\n Context: {context}\n Explanation: """

major_prompt = """From the perspective of the syllogism, please propose a major premise for the question according to the context and the explanation. In a syllogism, the major premise is a general statement or a universal truth. Note that the major premise must be found or supported in the context and as much as possible related to the question.

Question: {question}\n Context: {context}\n Explanation: {explanation}\n Major Premise: """

minor_question_prompt = """From the perspective of the syllogism, please ask a minor premise question for the given question. The minor premise question you ask should be as relevant as possible to the given question.

Question: {question}\n Context: {context}\n Major Premise: {major}\n Minor Premise Question: """

minor_prompt = """Please according to the context, think step by step to answer the question. Note that You must think and answer only based on the context.

Question: {minor_question}\n Context: {context}\n Minor Premise: """

final_prompt = """From the perspective of the syllogism, please according to the major premise and minor premise, think step by step to answer the question.

Major Premise: {major}\n Minor Premise: {minor}\n Question: {question}\n Answer: """

Figure 3: Prompts for each stage of our SR-FoT.

clude SC-CoT, C-CoT and SC-SR-FoT, we perform 10 samplings each. To enhance the diversity of sampling outcomes, the hyperparameters on GPT-3.5-turbo and Qwen1.5-32B-Chat for top_p and temperature are adjusted to 0.7 and 0.9 respectively. The temperature hyperparameter on DeepSeek remained at the default recommended value of 1. The number of in-context example prompts used in all the methods on the ScienceQA, StrategyQA, and BoolQ datasets are 5, 2, and 2, respectively.

Experimental Results and Analyses

Performance on ScienceQA. Scientific question answering is a task scenario that often requires deductive reasoning. As seen in Tab. 1, under GPT-3.5-turbo, in the comparison of single-round sampling methods, our SR-FoT outperforms the Base and CoT methods by 1.5% and 0.5% respectively and is on par with multi-round sampling methods like SC-CoT and C-CoT. In the comparison of multi-round sampling methods, our SC-SR-FoT exceeds SC-CoT and C-CoT methods by 1.5%. Under the open-source model DeepSeek-V2, SR-FoT outperforms the Base and CoT by 4.8% and 3.2% respectively, even surpassing multi-round sampling methods. What's more, our SC-SR-FoT further increases the accuracy to 93.0%. Under Qwen1.5-32B-Chat, compared to the Base and CoT methods, our SR-FoT has an improvement of 1.3% and 1.6% respectively. Compared to SC-CoT and C-CoT, our SC-SR-FoT also performs better, surpassing them by 2.8% and 3.2% respectively. These indicate that our methods achieves greater superiority on the ScienceQA dataset under multiple models.

Performance on StrategyQA and BoolQ. StrategyQA and BoolQ are two other knowledge-based reasoning question-answer datasets that require a true or false judgment based on context or common sense knowledge. From Table 1, for StrategyQA under GPT-3.5-turbo, in the comparison of single-round sampling methods, our SR-FoT outperforms Base and CoT by 7.3% and 0.8% respectively; in the comparison of multi-round sampling methods, our SC-SR-FoT exceeds SC-CoT and C-CoT by 1.9% and 3.2% respectively. Similar trends are observed under DeepSeek-V2 and Qwen1.5-32B-Chat. In addition, our SR-FoT and SC-SR-FoT also perform the best in both single-round sampling methods and multi-round sampling methods on BoolQ under the three models.

Overall, whether under the closed-source large language model GPT-3.5-turbo or the open-source large language models DeepSeek-V2 and Qwen1.5-32B-Chat, our SR-FoT achieve a superior accuracy compared with other compared methods on the ScienceQA, StrategyQA, and BoolQ datasets. This demonstrates the effectiveness of our SR-FoT.

It is worth noting that under DeepSeek-V2 and Qwen1.5-32B-Chat, the Base method achieves relatively high results across all three datasets, while the benefits of the CoT method show signs of saturation, and at times perform worse than the Base method. However, our methods, whether under single-round sampling settings (SR-FoT) or multi-round aggregated sampling settings (SC-SR-FoT), are still able to further enhance performance, demonstrating greater potential for performance gains. We believe this is because our SR-FoT employs a syllogistic deductive reasoning frame-

Question: Which word does not rhyme? **Options**: ['bib', 'job', 'sob'] **Ground Truth**: bib

CoT:

The answer is 'job', reasoning process step by step:

1.The question asks which word does not rhyme with the others.

2.The options provided are 'bib', 'job', and 'sob'.

3.To determine which word does not rhyme, we need to consider the ending sounds of each word.

4.'Bib' and 'sob' both end with the same 'b' sound, making them rhyme with each other.

5.However, 'job' ends with a different sound, the 'b' sound, making it not rhyme with 'bib' and 'sob'.

6.Therefore, based on the context and the ending sounds of the words, 'job' does not rhyme with the others.

SR-FOT:

Question Explanation:

The question is asking which word does not rhyme with the others. To answer the question, we need to understand the concept of rhyming words and compare the sounds of each word in the options to determine which one does not rhyme.

Major Premise Production:

Rhyming words have similar ending sounds, typically the vowel and consonant sounds in the final syllables of the words.

Posing the Minor Premise Question:

Which word has a different ending sound compared to the others?

Minor Premise Production:

The word 'bib' has a different ending sound compared to the others.

Final Syllogistic Reasoning:

The answer is 'bib'. The reasoning process of syllogism is as follows: Since rhyming words have similar ending sounds, typically the vowel and consonant sounds in the final syllables of the words, and the word 'bib' has a different ending sound compared to the others, it can be concluded that 'bib' does not rhyme with 'job' and 'sob'.

Figure 4: A case of using CoT and SR-FoT to answer a question in the ScienceQA dataset respectively. The highlighted red parts indicate the incorrect or misleading content, while the highlighted green parts indicate the content that helps correct reasoning.

work, allowing LLMs to address these knowledge-based reasoning tasks based on a more rigorous reasoning process, thereby achieving better overall performance.

Ablation Study

Effectiveness Comparisons for Subcategories. As shown in table 2, we conduct the experiments on the ScienceQA dataset with DeepSeek-v2. The results demonstrate that our methods can enhance the reasoning performance of the language model across questions of different difficulty levels and various subjects, compared with the state-of-the-art methods. When increasing the consistency of the proposed method, the beneficial effects become more significant.

Ablation of Stages. As shown in Table 3, we conduct experiments on ScienceQA under DeepSeek-V2 to verify the effectiveness of each stage in our method. Specifically, 'all in one stage' denotes using instructions and examples to let the LLM provide the premises based on the question and options, and then directly provide the answers. 'w/o stage 3' denotes providing the minor premise directly, instead of posing it as a question first and then answering. The results demonstrate that the completeness of each stage is important. In detail, discarding either the problem explanation or the major and minor premises would decrease the performance. Furthermore, allowing the language model to directly provide the major and minor premise would significantly reduce its performance, demonstrating the necessity of the multi-stage thinking framework in our SR-FoT.

Impact of Visible Information in Various Stages. As shown in Table 4, we conduct the experiments on the StrategyQA dataset with DeepSeek-v2. 'w/o context in stage 3' denotes that the minor premise question is generated without considering the context. 'add Q_{ori} in stage 4' denotes that providing the original question, minor premise question, and context all to the LLM during the process of answering the minor premise question. The results demonstrate that both decreasing or increasing the content of the input prompts adversely affect performance. This underlines the appropriateness of the designed visible information at each stage of our SR-FoT.

Rigor Analysis

To more directly analyze whether our SR-FoT improves the rigor of the reasoning process compared to CoT, we randomly select 50 cases from each of the three datasets under GPT-3.5-turbo for manual evaluation. For CoT and SR-FoT, if all intermediate steps from the first step of reasoning to obtaining the final answer are correct and logically progressive, without any factual inconsistencies or self-inconsistencies, we call them rigorous; otherwise, they are not rigorous. The results are in Table 5. From the table, it can be found that